SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Autonomous Smart Caddy Cart (ASCC)

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Submitted by Group 07

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Acronyms

1. Introduction

1.1 Motivation and Problem

Golfers who prefer walking the course must transport their clubs and equipment between shots, either by carrying a golf bag or manually operating a push cart. While conventional electric push carts reduce the physical effort required to move equipment, many still require direct user control. Commercial autonomous carts provide golfer-following capabilities, but these systems primarily address equipment transportation rather than the broader challenges of managing golf clubs and accessing useful information during a round.

An additional challenge is ensuring that all clubs are accounted for throughout play. Golfers frequently remove clubs from their bags at different locations, creating opportunities for equipment to be misplaced or left behind. Existing solutions for autonomous transportation and golf equipment management are often implemented separately, requiring golfers to rely on multiple devices or manual checks.

The Autonomous Smart Caddy Cart (ASCC) aims to address these limitations by combining autonomous equipment transportation with club identification and tracking in a single system. The project focuses on developing a practical prototype that reduces the need for manual cart operation while helping golfers manage their equipment.

1.2 Project Objectives

The primary objective of the ASCC is to develop a motorized golf push cart capable of following a golfer while maintaining safe operation in a controlled outdoor environment. The system will use a controller-based following mechanism rather than computer vision, reducing the computational complexity associated with visual target detection and tracking.

The cart will incorporate distance sensors to detect obstacles and support automatic stopping or avoidance. An independent emergency-stop mechanism will provide an additional means of preventing unintended motion. In parallel, an RFID-based club-tracking subsystem will identify tagged golf clubs and determine whether expected clubs are present in the bag.

The completed prototype is intended to demonstrate the integration of these functions into a single platform, with emphasis on reliable subsystem operation, safe movement, and clear communication of equipment status to the user.

1.3 Project Scope

The project will prioritize motorized movement, golfer following, obstacle detection, emergency stopping, and RFID-based club identification. These functions form the minimum viable system and will guide the design requirements, subsystem prototypes, and verification and validation procedures.

Additional functionality, such as collecting shot data, recommending clubs, or providing a mobile application, may be considered if time and resources permit. These features are excluded from the core project scope because they introduce additional sensing, software, and interface requirements that are not essential to demonstrating the primary system objectives.

The ASCC is intended as a prototype rather than a commercially deployable product. Testing will focus on controlled indoor environments followed by representative outdoor conditions. Operation on public roads, highly congested areas, and extreme terrain is outside the intended scope.

1.4 System Overview and Operating Scenario

The ASCC consists of a motorized push-cart platform, an onboard computing system, motor-control electronics, distance sensors, a golfer-following interface, and an RFID-based club-tracking subsystem. The Raspberry Pi will coordinate high-level system functions, while the motor-control electronics will regulate movement and respond to safety-related inputs.

During normal operation, the golfer activates the system and begins walking along the course. The cart receives information from the following mechanism and adjusts its movement to maintain an appropriate distance from the golfer. Distance sensors monitor the surrounding area for obstacles, allowing the control system to stop or modify movement when an obstruction is detected. The golfer can also activate an emergency stop when immediate intervention is required.

When clubs are returned to or removed from the bag, the RFID subsystem attempts to identify their presence. The system maintains a record of detected clubs and can notify the golfer when an expected club is missing.

The system boundary includes the cart, its onboard electronics, sensors, power supply, and club-tracking hardware. External elements include the golfer, the following controller, tagged clubs, obstacles, and the operating terrain. The design assumes a suitable walking surface, adequate battery charge, and environmental conditions within the operating limits of the selected hardware.

1.5 Design Constraints

The ASCC must be developed within the project budget of CAD $600 and the available capstone development schedule. The design therefore emphasizes reuse of existing components, including the Raspberry Pi and ultrasonic sensors, together with a second-hand motorized push-cart platform.

Hardware compatibility, sensor reliability, battery capacity, and safe motor operation will influence the final design. Particular attention will be given to detecting obstacles, preventing unintended movement, and ensuring that the cart can be stopped reliably.

These constraints establish the basis for the functional decomposition, requirements specifications, development approach, and verification and validation procedures presented in the following sections.

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2.1 System Overview and Design Alternatives

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7. Resource Requirements

The Autonomous Smart Caddy Cart (ASCC) will be developed using a combination of existing laboratory equipment, previously acquired components, and purchased hardware. Resource selection is guided by the system requirements, subsystem dependencies, and the project budget of CAD $600. The team will prioritize components needed for motor control, golfer following, obstacle detection, emergency stopping, and RFID-based club identification before acquiring hardware for additional features.

7.1 Hardware and Equipment

The Raspberry Pi 5 will serve as the primary computing platform for sensor processing and high-level control. A motor controller will interface with the drive system of a used motorized golf push cart, which provides the mechanical platform for the prototype. Ultrasonic sensors and a TF-Luna LiDAR sensor will support obstacle detection and distance measurement, while RFID readers and tags will support club identification.

Table 1 summarizes the estimated hardware requirements. Components already available to the team are assigned no additional purchase cost.

Table 1: Estimated ASCC bill of materials (CAD).

Component	Quantity	Cost (CAD)
Raspberry Pi 5 (existing)	1	$0
Ultrasonic sensors (existing)	Multiple	$0
RFID readers and tags	2 sets	$20
TF-Luna LiDAR sensor	1	$50
Used motorized golf push cart	1	$250
Motor controller	1	$50
Battery and power supply	1 set	$75
Mounting hardware	Assorted	$30
Wiring, connectors, and switches	Assorted	$30
Estimated tax and shipping	–	$75
Contingency allowance	–	$15
Total estimated cost		**$595**

The motorized push cart represents the largest anticipated expenditure and will be sourced second-hand to reduce the cost of mechanical fabrication. The selected cart must be compatible with the proposed motor-control architecture and provide sufficient space for the battery, computing hardware, sensors, and club-tracking electronics. The

final bill of materials will be updated with manufacturer or supplier references, confirmed quantities, and actual purchase prices.

7.2 Resource Allocation and Milestone Dependencies

Resources will be acquired and integrated according to the order of subsystem development. Table 2 identifies the major resource dependencies and the development activities they support. Milestone identifiers will be aligned with the final project schedule.

Table 2: Resource requirements and development dependencies.

Development activity	Required resources	Dependency
Motor control prototype	Cart, motor controller, battery	Mechanical platform and power compatibility
Sensor integration	Raspberry Pi, ultrasonic sensors, LiDAR	Stable power and sensor interfaces
Golfer-following prototype	Controller, computing platform, drive system	Motor-control prototype
Club identification prototype	RFID readers, tagged clubs	Reliable tag detection and reader placement
Obstacle avoidance and emergency stop	Distance sensors, motor controller, stop switch	Sensor and motor-control integration
System integration	Mounting hardware, wiring, connectors	Individual subsystem verification
Outdoor validation	Completed cart, test equipment, suitable test area	Integrated prototype and safety checks

The team will verify electrical compatibility, mechanical fit, and interface requirements before finalizing purchases. Components with longer procurement times, particularly the used cart and specialized sensors, will be prioritized to avoid delaying dependent milestones.

7.3 Development Facilities and Software

The project will require access to electronics workspaces, soldering equipment, multimeters, power supplies, and mechanical assembly tools for subsystem prototyping and integration. Software development will use Python and the Raspberry Pi development environment, with version control used to coordinate changes across team members.

Initial testing will take place in a controlled indoor environment where obstacle placement, operating speed, and emergency-stop behaviour can be evaluated safely. Outdoor testing

will require a suitable area with permission for operation, sufficient space for the cart, and representative terrain conditions. Access to facilities and test equipment will be confirmed before the corresponding validation milestones.

7.4 Budget and Procurement Contingencies

The estimated total project cost is CAD $595, leaving CAD $5 below the project budget limit of CAD $600. The estimate includes a $15 contingency allowance, but the small remaining margin makes the project sensitive to changes in cart pricing, shipping costs, and component availability.

If the selected cart exceeds the budget or is unavailable, the team will evaluate alternative second-hand platforms before committing to additional purchases. Existing components and laboratory equipment will be reused wherever practical. Optional features will be deferred if necessary to preserve resources for the core following, obstacle detection, emergency-stop, and club-tracking functions.

8. Conclusion

9. References

A. Detailed Requirements and Traceability

B. Supporting Technical Calculations

C. Detailed Test Procedures

D. Detailed Resource and Budget Information